

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0398

Kyocera 138 kV
December 22, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0398. The Interconnection Customer proposes a 160.9 MW wind facility interconnecting at the Kyocera 138 kV substation in the Virginia area. The request was received on October 31, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 237 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,084,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Riverview – Comox 138 KV Line is overloaded to 129.1% of its 230 MVA emergency rating for the loss of the Comox – Ruch 138 KV. The project contributes 100.0% DFAX (160.9 MW) to this overload.

Comox – Ruch 138 KV L: 116.5% loading (255 MVA) following the outage of the Riverview Substation – Comox 138 KV; DFAX = 100.0%.

The Bearskin – Ruch 138 KV is overloaded to 115.4% of its 253 MVA emergency rating for the loss of the Comox – Riverview 138 KV Line. The project contributes 100.0% DFAX (160.9 MW) to this overload.

For the loss of Comox – Riverview 138 KV line, the Chatham – Bearskin 138 KV loads to 110.3% of its 243 MVA rating. Project DFAX: 100.0%.

Lyondell – Chatham 138 KV: 107.8% loading (240 MVA) following the outage of the Riverview – Comox 138 KV line; DFAX = 100.0%.

Rustic Substation – Hughson 345 KV: 105.6% loading (1,268 MVA) following the outage of the Dixleta – Rustic 345 KV; DFAX = 12.3%.

Tombstone – Bixby Hill 500 KV: 102.9% loading (3,188 MVA) following the outage of the Tombstone – Sharyland 345 KV; DFAX = 8.6%.

For the loss of Louisa Generation Facility – N Anna 345 KV, the Louisa Generation Facility – Carrabelle 345 KV loads to 100.9% of its 1,562 MVA rating. Project DFAX: 11.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Riverview – Comox 138 KV L	Rebuild line with higher-capacity conductor	\$2,008,000	100% (\$2,008,000)
Comox – Ruch 138 KV	Rebuild line with higher-capacity conductor	\$13,855,000	100% (\$13,855,000)
Bearskin – Ruch 138 KV Line	Rebuild line with higher-capacity conductor	\$16,075,000	100% (\$16,075,000)
Chatham – Bearskin 138 KV	Rebuild line with higher-capacity conductor	\$4,583,000	100% (\$4,583,000)
Chatham – Lyondell 138 KV	Reconductor line	\$3,655,000	100% (\$3,655,000)
Rustic – Hughson 345 KV	Reconductor line	\$15,837,000	3% (\$525,000)
Tombstone – Bixby Hill 500 KV	Reconductor line	\$63,387,000	3% (\$1,890,000)
Louisa Generation Facility – Carrabelle 345 KV	Reconductor line	\$49,758,000	3% (\$1,493,000)
	Total allocated to project		\$44,084,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.